Portfolio Selection Models Considering Fuzzy Preference Relations of Decision Makers

Weiwei Guo, Wei-Guo Zhang, and Xiaoqing Chen

***Abstract*—During the portfolio selection process, investors often have individual behavioral preferences that may be fuzzy, uncertain, and incomplete. For the first time, this article combines fuzzy preference relations (FPRs) with portfolio theory to study the portfolio selection problem faced by investors. First, we improve the relations between judgment elements and priority vectors in FPRs and develop a new definition of additively consistent FPRs under priority vectors. On the basis of this and Markowitz portfolio models, we then propose two portfolio models considering FPRs with unknown preference information and three types of portfolio models considering FPR with partially known preference information. We conduct a comparative analysis of these models and derive a flowchart of investment selection under different conditions. Finally, we use an empirical example to compare the results of all models and analyze their sensitivity to various parameters. The results demonstrate that increasing the distance threshold between judgment elements has a positive effect on the objective function of portfolio models considering FPRs, although this effect gradually becomes saturated. Additionally, the models proposed in this article exhibit high robustness to the consistency index of FPRs.**

***Index Terms*—Additive consistency, fuzzy preference relation (FPR), mean–variance model, portfolio selection.**

I. Introduction

In FINANCIAL markets, the trading of numerous securities is conducted by investors, agents, fund managers, and other individuals or groups in the pursuit of profits [1], [2], [3], [4], [5]. In recent years, portfolio models and methodologies have been closely linked to other theories, exhibiting their multidimensional trends and increasingly significant practical role [6], [7], [8], [9]. Portfolio models have been investigated under linear programming, objective programming, multiobjective programming, dynamic programming, robust optimization, and multistage and multiattribute (criteria) [10], [11], [12], [13], [14], [15]. Moreover, portfolio models based on different theories, such as probability theory, possibility theory, credibility theory, fuzzy theory, and uncertainty theory, have also been proposed, in which variables, such as return and risk, are transformed from traditional crisp values to interval numbers, random variables, uncertain variables, and fuzzy numbers [16], [17], [18], [19]. For example, Qin [16] constructed a mean–variance model with both stochastic and uncertain returns based on uncertainty theory. At the intersection with other fields, portfolio theory is closely associated with data envelopment analysis, utility theory, and prospect theory, which extends portfolio models, methods, and applications [20], [21], [22], [23]. In actual applications, mean–variance models and portfolio theory are widely used in financial investment, risk analysis, comprehensive evaluation, and predictive decisions [24], [25], [26].

During the investment process, investors typically wish to retain their own preferences for different securities. Thus, the question of how to make scientific and reasonable investments while retaining the investor's preference information has become an issue of interest. Fuzzy preference relations (FPRs) are an effective means of dealing with the uncertain preferences of decision makers. More importantly, the priority vectors corresponding to FPRs that satisfy an additive consistency criterion can be integrated with investment vectors. Therefore, this article describes the use of FPRs to construct portfolio models that account for investor preferences. FPRs can be used to form a judgment matrix, with a scale of 0–1 applied to characterize the decision maker's preferences for different alternatives [27], [28], [29], [30]. FPRs have been expanded to interval FPRs, intuitionistic FPRs, hesitating FPRs, and uncertain preference relations [31], [32], [33], [34], [35], [36]. Consistency is the most important indicator of an FPR, determining the logic and rationality of the individual's ranking of different alternatives [37], [38], [39], [40]. Tanino [28] defined the additive consistency and multiplicative consistency of FPRs, while Herrera-Viedma et al. [41] and Li et al. [42] derived expressions for calculating consistency indices of FPRs. Currently, FPRs are widely used in decision analysis, solution ranking, and comprehensive evaluation [43], [44].

In current portfolio selection, it is typically assumed that investors are fully rational individuals with no preference differences toward investment objects, aiming solely for risk minimization and profit maximization. However, in the

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real-world investment process, investors often exhibit varying degrees of preference differences due to differences in their cognition, experience, and psychology. For instance, risk-tolerant investors tend to favor high-risk, high-return securities, while risk-averse investors prefer low-risk, low-return securities. Therefore, there is a need for a method that can capture these preference differences while integrating well with the weight in portfolio selection. According to the analysis of this article, FPRs perfectly meet these requirements, making the introduction of FPRs into portfolio selection necessary and reasonable. In addition, the introduction of FPRs into portfolio selection for the first time can combine portfolio theory and preference relations well, which provides a new approach to behavioral finance and also expands the application of FPRs in actual decision making.

The main advantages of introducing FPRs into portfolio selection compared with traditional methods are as follows.

1) The derivation of preferences by pairwise comparison in FPRs is more scientific, intuitive, and easier to implement: FPRs use the pairwise comparison method to formulate judgment results for all securities in turn, whereby the judgment element is employed as the preference for different securities. When the number of securities is large, the scientific rationality and implementation of this method are superior to traditional methods because it is unrealistic for investors to directly assign accurate weights or preferences, but it is feasible and highly convenient to assign preferences through pairwise comparisons.
2) Because all judgment elements in FPRs are independent, when the value of a judgment element changes, other elements are unaffected. However, in traditional methods, when the weight of a security increases or decreases, the weights of other securities are inevitably affected, and determining the changes in the weights of other securities is difficult.
3) FPRs that completely satisfy the consistency criteria have a unique priority vector that can be perfectly combined with an investment strategy. Thus, FPRs can deal with portfolio selection problems with preferences.

On the basis of these advantages of using FPRs to solve portfolio selection problems with preferences, we summarize the following improvements in existing portfolio models, as well as the challenges of introducing FPRs into portfolio selection.

1) Existing portfolio models do not account for the FPRs of investors for different securities before investment.
2) The completeness and consistency of investors' preference information and preference relations are not considered.
3) The relations between the investment vectors and priority vectors of FPRs have not been investigated.
4) Models for selecting a portfolio while maximizing the retention of the original investor preference information have not previously been constructed.

Considering the completeness and consistency of the investor FPRs, this article proposes some novel portfolio models in which FPRs integrate the uncertain preference information of investors and links are constructed between FPRs and mean–variance models through priority vectors. These models aim to maximize the consistency index of FPRs and optimize the investment strategy while retaining as much of the original preference information of investors as possible. The main innovations of this article are as follows.

1) This is the first attempt to introduce FPRs into portfolio theory through priority vectors, providing a theoretical basis for the introduction of interval FPRs, intuitionistic FPRs, and other fuzzy constructs in the future.
2) The relations between the judgment elements and the priority vector in the FPRs are improved, and the priority vectors are used to link FPRs to the mean–variance model.
3) Two portfolio models considering FPRs with unknown preference information and three portfolio models considering FPRs with partially known preference information are proposed. A comparative analysis of these models is presented, and a portfolio flowchart for different environments is developed.
4) The portfolio selection process is implemented to maximize the retention of the investor's original preference information, and a method is derived for modifying the investor's preference information to achieve better investment results.

The remainder of this article is structured as follows. Section II introduces the basics of FPRs and portfolio theory. Section III improves the relations between the judgment elements and the priority vectors in FPRs and gives a new definition of the additive consistency of FPRs under these priority vectors. In Section IV, two types of portfolio models considering FPRs with unknown preference information and three types of portfolio models considering FPRs with partially known preference information are proposed. Comparative analysis of these models and a portfolio flowchart for different circumstances are provided in Section V. In Section VI, an empirical example is presented to compare the results of the different models, and the sensitivity of the models to various parameters is analyzed. Finally, Section VII outlines the conclusions of this study and some ideas for future work.

II. Preliminaries

In this section, we introduce the basics of FPRs and portfolio theory. For simplicity, we define $X = \{x_1, x_2, \dots, x_n\}$ as a finite set of alternatives, where x_i denotes the ith alternative, $i \in \{1, 2, \dots, n\} = N$.

A. Fuzzy Preference Relations

Definition 1 (See [27], [28], [29], [30]): An FPR on X is denoted by a square matrix $P = (p_{ij})_{n\times n} \subset X \times X$, where $p_{ij} + p_{ji} = 1, p_{ii} = 0.5, p_{ij} \geq 0, i, j \in N$.

The element p_{ij} in FPR P expresses the preference degree for alternative x_i over alternative x_j. A value of $p_{ij} = 0.5$ indicates that there is no preference between x_i and x_j; if $0.5 < p_{ij} \leq 1$, then x_i is preferred to x_j, and if $0 \leq p_{ij} < 0.5$, then x_j is preferred to x_i.

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Definition 2 (See [28]): An FPR $P = (p_{ij})_{n\times n}$ has the property of additive consistency if

$$p_{ij} + 0.5 = p_{ik} + p_{kj},\ \ i, j, k \in N. \tag{1}$$

Based on (1), whether p_{ij} satisfies this consistency property depends on the value of $p_{ik} + p_{kj} - 0.5,\ \ i, j, k \in N$. The smaller the deviation of p_{ij} from $p_{ik} + p_{kj} - 0.5$, the higher the degree of consistency of p_{ij}. In this regard, the following formula is proposed for determining the consistency index of FPR [41], [42].

Definition 3 (See [41], [42]): The consistency index of an FPR P is

$$CI(P) = 1 - \frac{2}{3n(n-1)(n-2)} \sum_{i,j,k=1, i\neq j\neq k}^{n} |p_{ik} + p_{kj} - p_{ij} - 0.5|. \tag{2}$$

Obviously, the range of $CI(P)$ is the interval [0, 1]. Larger values of $CI(P)$ correspond to a higher degree of additive consistency of FPR P. When $CI(P) = 1$, P is regarded as completely consistent.

Definition 4 (See [28]): An FPR $P = (p_{ij})_{n\times n}$ is additively consistent if and only if there exists a non-negative vector $w = (w_1, w_2, \ldots, w_n)$, $|w_i - w_j| \le 1$, $i, j \in N$ such that

$$p_{ij} = 0.5\big(w_i - w_j + 1\big), \quad i, j \in N. \tag{3}$$

The priority vector $w = (w_1, w_2, \ldots, w_n)$ represents the degree of importance of the set of alternatives $X = \{x_1, x_2, \ldots, x_n\}$. However, Definition 4 does not guarantee that the sum of the weights is 1. Therefore, the following definition of the relationship between the element p_{ij} and w_i, w_j in FPR P is proposed.

Definition 5 (See [45], [46], [47], [48]): An FPR $P = (p_{ij})_{n\times n}$ is additively consistent if there exists a non-negative vector $w = (w_1, w_2, \ldots, w_n)$ that satisfies $\sum_{i=1}^{n} w_i = 1, i \in N$, such that

$$p_{ij} = 0.5\big(w_i - w_j + 1\big), \quad i, j \in N. \tag{4}$$

The difference between Definitions 2, 4, and 5 is that Definition 2 defines the additive consistency of the FPR in terms of the relationship between judgment elements, whereas Definitions 4 and 5 define the additive consistency of the FPR in terms of the relationship between judgment elements and the priority vector. The latter formulation accurately corresponds to investment strategies, allowing us to relate FPRs to portfolio selection. Equation (1) must hold when p_{ij} satisfies (3) or (4).

Definition 5 gives the relations between p_{ij} and w_i, w_j, which satisfy $\sum_{i=1}^{n} w_i = 1$. However, it is not exactly equivalent to Definition 2. Moreover, when FPR P satisfies additive consistency under Definition 2, it is not always possible to find the corresponding priority vector under Definition 5. Accordingly, there is potential for improvement in Definitions 4 and 5, as discussed in Section III.

B. Portfolio Theory

We consider investors to be rational, meaning that they always want to make decisions that minimize risk for a given return or maximize return for a given risk. Let $r = (r_1, r_2, \ldots, r_n)$ and $\mu = (\mu_1, \mu_2, \ldots, \mu_n)$ be the returns and expected returns, respectively, of n securities, where r_i is uncertain and can be regarded as a random variable. Thus, μ_i is the expectation of r_i, that is, $\mu_i = E(r_i), i \in N$. Let the matrix $D(r) = (\sigma_{ij})_{n\times n}$ represent the covariance matrix of the random variable $r = (r_1, r_2, \ldots, r_n)$, and $w = (w_1, w_2, \ldots, w_n)$ be the weight vector of investing in n types of securities satisfying $\sum_{i=1}^{n} w_i = 1$. When $w_i \ge 0$, short selling is not allowed in the securities market. When w_i is negative, the market is being sold short. For convenience, this article only considers the case without short selling. On the basis of the above assumptions, Markowitz [49] constructed the following two types of models.

Assuming that the given expected rate of return is μ_0, the portfolio model that minimizes risk can be written as follows:

$$\begin{aligned} &\text{Min } \sum_{i=1}^{n}\sum_{j=1}^{n} w_i\sigma_{ij}w_j \\ &\text{s.t.} \begin{cases} \sum_{i=1}^{n} w_i\mu_i = \mu_0, & i \in N \\ \sum_{i=1}^{n} w_i = 1, & i \in N \\ w_i \ge 0, & i \in N. \end{cases} \end{aligned} \tag{5}$$

Assuming that the given investment risk is σ_0^2, the portfolio model that maximizes returns can be expressed as follows:

$$\begin{aligned} &\text{Max } \sum_{i=1}^{n} w_i\mu_i \\ &\text{s.t.} \begin{cases} \sum_{i=1}^{n}\sum_{j=1}^{n} w_i\sigma_{ij}w_j = \sigma_0^2, & i, j \in N \\ \sum_{i=1}^{n} w_i = 1, & i \in N \\ w_i \ge 0, & i \in N. \end{cases} \end{aligned} \tag{6}$$

III. New Definition of the Priority Vector

In Section II, we introduced the priority vector of an FPR and provided two different definitions: Definition 4 and Definition 5. However, both of these definitions have some deficiencies and can be further improved. This section analyzes these deficiencies and provides a new definition of the FPR using the priority vector.

Definition 4 cannot guarantee that $\sum_{i=1}^{n} w_i = 1$, and it is not possible to directly determine a priority vector that satisfies additive consistency. In contrast, Definition 5 guarantees $\sum_{i=1}^{n} w_i = 1$, but is not compatible with Definition 2, i.e., there are cases where the FPR satisfies additive consistency, but a suitable priority vector is not yielded under Definition 5. A detailed description is now given.

Theorem 1: The solution of w_i in the priority vector of the FPR under Definition 5 is

$$w_i = \frac{2\sum_{j=1}^{n} p_{ij} - n + 1}{n}, \quad i, j \in N. \tag{7}$$

Proof: Based on Definition 5, we obtain

$$\begin{aligned} p_{ij} &= 0.5\big(w_i - w_j + 1\big) \\ &\Rightarrow \sum_{j=1}^{n} p_{ij} = 0.5\left(nw_i - \sum_{j=1}^{n} w_j + n\right) \\ &\Rightarrow w_i = \frac{2\sum_{j=1}^{n} p_{ij} - n + 1}{n}. \end{aligned}$$

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However, when $2\sum_{j=1}^{n} p_{ij} - n + 1 < 0$, the result for w_i under (7) is negative, which contradicts the nonnegativity constraint on w_i in Definition 5. Although Definition 5 itself is completely correct, it is not capable of covering all FPRs satisfying additive consistency. In other words, when an FPR satisfies additive consistency under Definition 2, it may not be possible to find a priority vector that satisfies Definition 5. For example, the FPR

$$P = (p_{ij})_{n\times n} = \begin{pmatrix} 0.5 & 0.3 & 0.7 \\ 0.7 & 0.5 & 0.9 \\ 0.3 & 0.1 & 0.5 \end{pmatrix}$$

is additively consistent, but if the corresponding priority vector is solved using Definition 5, it yields $w_i = (0.333, 0.733, -0.067)$, in which one weight is negative. This contradicts the nonnegativity constraint on the weights in Definition 5. Therefore, it is necessary to redefine the relationship between p_{ij} and w_i, w_j so that any FPR satisfying additive consistency yields a corresponding compatible priority vector.

Definition 6: An FPR $P = (p_{ij})_{n\times n}$ satisfies additive consistency if there exists a non-negative vector $w = (w_1, w_2, \ldots, w_n)$ for which $\sum_{i=1}^{n} w_i = 1, i \in N$ such that

$$p_{ij} = \frac{n-1}{2}(w_i - w_j) + 0.5, \quad i, j \in N. \tag{8}$$

Theorem 2: The elements p_{ij} under (8) satisfy the definition of complementarity of symmetric elements in FPRs, as well as the additive consistency definition of FPRs. The range of w_i is [0, 1].

Proof: We prove the theorem in terms of the definition of complementarity of the symmetric elements in FPRs, the additive consistency definition, and the range of w_i.

1) In the definition of complementarity for FPRs

$$\begin{aligned} p_{ij} + p_{ji} &= \frac{n-1}{2}(w_i - w_j) + 0.5 + \frac{n-1}{2}(w_j - w_i) \\ &\quad + 0.5 = 1 \\ p_{ii} &= \frac{n-1}{2}(w_i - w_i) + 0.5 = 0.5 \end{aligned}$$

which is consistent with Definition 1, indicating that $p_{ij} = (n-1)/2(w_i - w_j) + 0.5$ reflects the complementary definition of FPRs.

2) In the definition of additive consistency for FPRs

$$\begin{aligned} p_{ij} + 0.5 &= \frac{n-1}{2}(w_i - w_j) + 0.5 + 0.5 \\ &= \frac{n-1}{2}(w_i - w_k) + 0.5 \\ &\quad + \frac{n-1}{2}(w_k - w_j) + 0.5 \\ &= p_{ik} + p_{kj} \end{aligned}$$

which is consistent with the additive consistency of FPRs (Definition 2), indicating that when $p_{ij} = (n-1)/2(w_i - w_j) + 0.5$, the FPR $P = (p_{ij})_{n\times n}$ must be additively consistent under Definition 2.

3) For the range of w_i, according to (8) and $\sum_{i=1}^{n} w_i = 1$, we obtain

$$\begin{aligned} p_{ij} &= \frac{n-1}{2}(w_i - w_j) + 0.5 \\ &\Rightarrow \sum_{j=1}^{n} p_{ij} = \frac{n-1}{2}\left(nw_i - \sum_{j=1}^{n} w_j\right) + 0.5n \\ &\Rightarrow w_i = \frac{1}{n}\left[1 + \frac{2}{n-1}\left(\sum_{j=1}^{n} p_{ij} - 0.5n\right)\right]. \end{aligned} \tag{9}$$

At this point, we have a method for calculating the priority vector for the FPR when additive consistency is completely satisfied (or the consistency index is 1). However, when the FPR does not completely satisfy additive consistency or the consistency index is less than 1, the FPR does not have a priority vector. On the basis of (9), when $p_{ij} = 1, j \in N, j \neq i$, and $p_{ii} = 0.5$, w_i attains its maximum value and we obtain

$$\text{Max } w_i = \frac{1}{n}\left[1 + \frac{2}{n-1}(0.5 + (n-1) - 0.5n)\right] = \frac{2}{n}.$$

Because the FPR is at least a second-order square matrix (that is, $n \geq 2$), the maximum value of w_i is less than or equal to 1. When $p_{ij} = 0, j \in N, j \neq i$, and $p_{ii} = 0.5$, w_i attains its minimum value and we obtain

$$\text{Min } w_i = \frac{1}{n}\left[1 + \frac{2}{n-1}(0.5 - 0.5n)\right] = 0.$$

Therefore, when $p_{ij} = (n-1/2)(w_i - w_j) + 0.5, \forall i, j \in N$, $0 \leq w_i \leq 1$ always holds. This demonstrates that, for any additively consistent FPR satisfying Definition 2, the corresponding priority vector under Definition 6 can always be found and satisfies $\sum_{i=1}^{n} w_i = 1, w_i \geq 0$. In addition, when $p_{ij} = 0.5, i, j \in N$, we have

$$w_i = \frac{1}{n}\left[1 + \frac{2}{n-1}(0.5n - 0.5n)\right] = \frac{1}{n}$$

which shows that when there is no difference among the alternatives, the weight of each alternative is $(1/n)$, which is consistent with the actual situation. ■

In summary, the formulation of Definition 6 provides perfect convergence with Definition 2, i.e., when $p_{ij} = (n-1/2)(w_i - w_j) + 0.5, i, j \in N$, the FPR will definitely satisfy additive consistency under Definition 2. Conversely, when the FPR satisfies additive consistency under Definition 2, there is guaranteed to be a priority vector satisfying Definition 6. Compared with the result $w_i = (0.333, 0.733, -0.067)$ obtained using (7), when we use (9) to calculate the priority vector, the result is $w = (0.333, 0.533, 0.133)$, which reflects the rationality and effectiveness of Definition 6.

IV. Portfolio Models With Preferences

A. Portfolio Models With Unknown Preference Information

In Sections II and III, we reviewed FPRs and portfolio theory and redefined the relations between judgment elements and priority vectors in FPRs. In this section, we integrate FPRs and portfolio models based on this theoretical background. Suppose that the decision maker has no preference for n securities, or that information about the decision maker's

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preferences is completely unknown. At the same time, we require the decision maker's FPRs for n securities under the optimal portfolio results. Naturally, the FPRs must satisfy additive consistency; otherwise, contradictory ranking results may be obtained.

On the basis of (5), (6), and (8), the following two types of models can be constructed.

Assuming that the expected rate of return is no less than μ_0, and that the investment risk is minimized, the following model ensures an additively consistent FPR for the decision maker:

$$\text{Min} \sum_{i=1}^{n}\sum_{j=1}^{n} w_i\sigma_{ij}w_j$$
$$\text{s.t.} \begin{cases} \sum_{i=1}^{n} w_i\mu_i \geq \mu_0, & i \in N \\ p_{ij} = \frac{n-1}{2}\left(w_i - w_j\right) + 0.5, & i,j \in N \\ \sum_{i=1}^{n} w_i = 1, & i \in N \\ p_{ij}, w_i \geq 0, & i,j \in N. \end{cases} \tag{10}$$

Compared with (5), we have imposed the constraint $p_{ij} = (n-1)/2(w_i - w_j) + 0.5, i,j \in N$, which ensures an additively consistent FPR under the weights of the optimal portfolio at the minimum risk. Additionally, the constraint itself implies that $p_{ij} + p_{ji} = 1$ and $p_{ii} = 0.5$. Thus, these additional constraints are not required, similar to (11).

Assuming that the investment risk is no higher than σ_0^2, and the expected return is maximized, the following model ensures an additively consistent FPR for the decision maker:

$$\text{Max} \sum_{i=1}^{n} w_i\mu_i$$
$$\text{s.t.} \begin{cases} \sum_{i=1}^{n}\sum_{j=1}^{n} w_i\sigma_{ij}w_j \leq \sigma_0^2, & i,j \in N \\ p_{ij} = \frac{n-1}{2}\left(w_i - w_j\right) + 0.5, & i,j \in N \\ \sum_{i=1}^{n} w_i = 1, & i \in N \\ p_{ij}, w_i \geq 0, & i,j \in N. \end{cases} \tag{11}$$

In a similar vein, unlike (6), we impose the constraint $p_{ij} = (n - 1/2)(w_i - w_j) + 0.5, i,j \in N$, which ensures that the FPR satisfies additive consistency under the weights of the optimal portfolio when returns are maximized.

Models (10) and (11) have only one more constraint than (5) and (6), respectively, and so their strategies for solving the same investment problem may be identical. However, the significance of (10) and (11) is that they obtain FPRs that fully satisfy the additive consistency corresponding to the optimal portfolio strategy. Thus, investors can compare the FPRs with their own inward or inherent preferences, stabilize their rational preferences and correct their irrational preferences, and provide a preference basis and reference for subsequent investments of the same type.

B. Portfolio Models Under Incomplete Preference Information

In Sections II and IV-A, we introduced classical portfolio models and portfolio models without preferences, both of which assume that the decision maker is entirely rational, i.e., seeks to minimize risk or maximize returns. However, in actual management decision-making problems, the decision maker is a significant part of the system, and their attitude, preferences, and utility in facing different decision-making problems will have a nonnegligible impact on the results. This is essentially behavioral operations theory, which is derived from traditional operations theory. In practical decision-making cases, decision makers are often influenced by their environment, preferences, experience, and other factors, resulting in unique information that is often incomplete, uncertain, or even contradictory. In the context of the portfolio problem considered in this article, we extend this to the idea that decision makers may provide their own FPRs for different securities in advance, depending on their own attitudes, experiences, and preferences combined with the investment background, environment, and market conditions at the time. These FPRs may be incomplete and inconsistent. In this regard, we need to consider the rate of return, risk, and consistency index of the FPRs over the course of the portfolio. Therefore, this article now considers the construction of portfolio models with incomplete preference information in terms of maximizing returns, minimizing risk, and maximizing the consistency index.

Prior to the construction of the models, we define an FPR $P^* = (p_{ij}^*)_{n\times n} \subset X \times X$, that is, P^* satisfies $p_{ij}^* + p_{ji}^* = 1$, $p_{ii}^* = 0.5$, $p_{ij}^* \geq 0$, $i,j \in N$, and $p_{ij}^* + 0.5 = p_{ik}^* + p_{kj}^*$, $i,j,k \in N$, with the aim of obtaining an FPR P^* that completely satisfies additive consistency and is closest to the FPR $P = (p_{ij})_{n\times n}$ given by the decision maker. Because the FPR $P = (p_{ij})_{n\times n}$ given by the decision maker contains only partial information and may not completely satisfy additive consistency, the accuracy and rationality of the FPR are made as high as possible by ensuring that the consistency index is no lower than a certain threshold α. Moreover, to make the FPR $P^* = (p_{ij}^*)_{n\times n}$ as similar as possible to $P = (p_{ij})_{n\times n}$, the distance between p_{ij} and p_{ij}^*, which completely satisfy additive consistency, should be no larger than the deviation threshold ε. On the basis of this, and in conjunction with (2), the following constraint is formulated:

$$\begin{cases} CI(P) \geq \alpha, & (12-1) \\ \left|p_{ij} - p_{ij}^*\right| \leq \varepsilon, \quad i,j \in N & (12-2) \end{cases} \tag{12}$$

where $CI(P) = 1 - 2/(3n(n-1)(n-2)) \sum_{i,j,k=1, i\neq j\neq k}^{n} |p_{ik} + p_{kj} - p_{ij} - 0.5|$. Constraint (12-1) guarantees that the consistency index of the FPR $P = (p_{ij})_{n\times n}$ is not lower than the threshold α from an overall perspective. A larger value of α gives a more reasonable FPR $P = (p_{ij})_{n\times n}$, and more accurate ranking results. Constraint (12-2) ensures that the distance between p_{ij} and p_{ij}^* is no greater than ε from each individual's perspective, so that the FPRs $P^* = (p_{ij}^*)_{n\times n}$ and $P = (p_{ij})_{n\times n}$ are identical or as consistent as possible. A smaller value of ε corresponds to a smaller gap between p_{ij} and p_{ij}^*, and preserves the original information of the investors to a greater extent. This not only ensures that the known information of the decision maker does not change significantly but also further improves the consistency index of the FPR given by the decision maker. Overall, (12-1) and (12-2) improve the consistency index of the FPR $P = (p_{ij})_{n\times n}$ while ensuring that the original preference information of the decision maker changes as little as possible.

In the following, portfolio models with incomplete preference information are constructed in terms of maximizing

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returns, minimizing risk, and maximizing the consistency index.

Assuming that the expected rate of return is no less than μ_0, and the consistency index of the FPR $P=(p_{ij})_{n\times n}$ provided by the decision maker is no less than α, the following model can be constructed to obtain an FPR $P^*=(p^*_{ij})_{n\times n}$ that satisfies additive consistency and minimizes the investment risk:

$$\begin{aligned}&\text{Min }\sum_{i=1}^{n}\sum_{j=1}^{n}w_i\sigma_{ij}w_j\\&\text{s.t.}\begin{cases}\sum_{i=1}^{n}w_i\mu_i\geq\mu_0, & i\in N\\ p^*_{ij}=\frac{n-1}{2}\left(w_i-w_j\right)+0.5, & i,j\in N\\ \sum_{i=1}^{n}w_i=1, & i\in N\\ CI(P)\geq\alpha & \\ \left|p_{ij}-p^*_{ij}\right|\leq\varepsilon, & i,j\in N\\ p_{ij}+p_{ji}=1, & i,j\in N\\ p_{ij},\ p^*_{ij},\ w_i\geq 0, & i,j\in N.\end{cases}\end{aligned}\tag{13}$$

The first constraint in (13) ensures that the return on the decision maker's investment is no less than μ_0. The second constraint represents the relation between p^*_{ij} and $w_i,\ w_j$ under additive consistency, and the third constraint ensures that the sum of the weights is 1. The fourth and fifth constraints are generated from (12-1) and (12-2), respectively, and guarantee that the consistency index of FPR $P=(p_{ij})_{n\times n}$ is not lower than α and the distance between p_{ij} and p^*_{ij} is not larger than the deviation threshold ε, respectively. These constraints improve the consistency index of the incomplete FPR given by the decision maker without changing the decision maker's original preference information any more than necessary. Moreover, the sixth constraint is the definition of complementarity for the FPR $P=(p_{ij})_{n\times n}$ (i.e., Definition 1), because the FPR $P^*=(p^*_{ij})_{n\times n}$ in which $p^*_{ij}=(n-1/2)(w_i-w_j)+0.5$ directly implies that $p^*_{ij}+p^*_{ji}=1,\ p^*_{ii}=0.5$. Thus, we do not need to add these constraints to the FPR $P^*=(p^*_{ij})_{n\times n}$. The seventh constraint ensures that $p_{ij},\ p^*_{ij}$, and w_i are non-negative.

Assuming that the given investment risk is no higher than σ_0^2 and the consistency index of the FPR $P=(p_{ij})_{n\times n}$ provided by the decision maker is no lower than α, the following model is constructed to obtain an FPR $P^*=(p^*_{ij})_{n\times n}$ that satisfies additive consistency and maximizes the expected rate of return:

$$\begin{aligned}&\text{Max }\sum_{i=1}^{n}w_i\mu_i\\&\text{s.t.}\begin{cases}\sum_{i=1}^{n}\sum_{j=1}^{n}w_i\sigma_{ij}w_j\leq\sigma_0^2, & i,j\in N\\ p^*_{ij}=\frac{n-1}{2}\left(w_i-w_j\right)+0.5, & i,j\in N\\ \sum_{i=1}^{n}w_i=1, & i\in N\\ CI(P)\geq\alpha & \\ \left|p_{ij}-p^*_{ij}\right|\leq\varepsilon, & i,j\in N\\ p_{ij}+p_{ji}=1, & i,j\in N\\ p_{ij},\ p^*_{ij},\ w_i\geq 0, & i,j\in N.\end{cases}\end{aligned}\tag{14}$$

The fundamental difference between (14) and (13) is that (13) takes investment risk as the objective function and the expected rate of return as the constraint, whereas (14) is the opposite. Regarding the remaining constraints, (14) is consistent with the interpretation of (13).

Assuming that the expected rate of return is no lower than μ_0 and the investment risk is no higher than σ_0^2, the following model can be constructed to obtain an FPR $P^*=(p^*_{ij})_{n\times n}$ that satisfies additive consistency and maximizes the consistency index of the FPR $P=(p_{ij})_{n\times n}$ given by the decision maker:

$$\begin{aligned}&\text{Max }\alpha\\&\text{s.t.}\begin{cases}\sum_{i=1}^{n}w_i\mu_i\geq\mu_0, & i\in N\\ \sum_{i=1}^{n}\sum_{j=1}^{n}w_i\sigma_{ij}w_j\leq\sigma_0^2, & i,j\in N\\ p^*_{ij}=\frac{n-1}{2}\left(w_i-w_j\right)+0.5, & i,j\in N\\ \sum_{i=1}^{n}w_i=1, & i\in N\\ CI(P)\geq\alpha, & \\ \left|p_{ij}-p^*_{ij}\right|\leq\varepsilon, & i,j\in N\\ p_{ij}+p_{ji}=1, & i,j\in N\\ p_{ij},\ p^*_{ij},\ w_i\geq 0, & i,j\in N.\end{cases}\end{aligned}\tag{15}$$

The fundamental difference between (15) and the previous two models is that the consistency index α is an unknown variable in (15), and the maximum value of α is obtained when the expected rate of return is not lower than μ_0 and the investment risk is not higher than σ_0^2. For the remaining constraints, (15) is consistent with (13).

Models (13)–(15) have two nonlinear constraints with absolute values, $CI(P)\geq\alpha$ and $|p_{ij}-p^*_{ij}|\leq\varepsilon,\ i,j\in N$. These need to be simplified to reduce the complexity of solving these models.

Theorem 3: Let $\alpha(i,k,j)_{t1}$ and $\alpha(i,k,j)_{t2},\ t=1,\ 2,\ldots,\ (3n(n-1)(n-2))/2$, be 0-1 variables representing the corresponding binary variables under the situation when $p_{ik}+p_{kj}-p_{ij}-0.5$. Then, the constraint that $CI(P)\geq\alpha$ becomes

$$\begin{cases}1-\frac{2}{3n(n-1)(n-2)}\sum_{i,j,k=1,i\neq j\neq k}^{n}\big[(\alpha(i,k,j)_{t2}\\ -\alpha(i,k,j)_{t1})*\left(p_{ik}+p_{kj}-p_{ij}-0.5\right)\big]\geq\alpha\\ p_{ik}+p_{kj}-p_{ij}-0.5\geq-1.5\alpha(i,k,j)_{t1}\\ p_{ik}+p_{kj}-p_{ij}-0.5\leq 1.5\alpha(i,k,j)_{t2}\\ \alpha(i,k,j)_{t1}+\alpha(i,k,j)_{t2}=1\\ \alpha(i,k,j)_{t1},\ \alpha(i,k,j)_{t2}\in\{0,1\}\end{cases}\tag{16}$$

where $t=1,\ 2,\ldots,\ (3n(n-1)(n-2))/2$.

Proof: First, because $0\leq p_{ij}\leq 1$, we have that $-1.5\leq p_{ik}+p_{kj}-p_{ij}-0.5\leq 1.5$. Second, because $\alpha(i,k,j)_{t1}+\alpha(i,k,j)_{t2}=1,\ \alpha(i,k,j)_{t2},\ \alpha(i,k,j)_{t2}\in\{0,1\}$. Thus, we need only consider two cases: $\alpha(i,k,j)_{t1}=1,\ \alpha(i,k,j)_{t2}=0$ and $\alpha(i,k,j)_{t1}=0,\ \alpha(i,k,j)_{t2}=1$. When $\alpha(i,k,j)_{t1}=1$, $\alpha(i,k,j)_{t2}=0$, we obtain $-1.5\leq p_{ik}+p_{kj}-p_{ij}-0.5\leq 0$, at which point $[\alpha(i,k,j)_{t2}-\alpha(i,k,j)_{t1}]*(p_{ik}+p_{kj}-p_{ij}-0.5)=-(p_{ik}+p_{kj}-p_{ij}-0.5)$, which is consistent with the result for $|p_{ik}+p_{kj}-p_{ij}-0.5|$ in this range. By analogy, when $\alpha(i,k,j)_{t1}=0,\ \alpha(i,k,j)_{t2}=1$, we obtain $0\leq p_{ik}+p_{kj}-p_{ij}-0.5\leq 1.5$, at which point $[\alpha(i,k,j)_{t2}-\alpha(i,k,j)_{t1}]*(p_{ik}+p_{kj}-p_{ij}-0.5)=(p_{ik}+p_{kj}-p_{ij}-0.5)$, which is consistent with the result for $|p_{ik}+p_{kj}-p_{ij}-0.5|$ in this range. Therefore, (16) is equivalent to $CI(P)\geq\alpha$. ■

In summary, it is sufficient to replace $CI(P)\geq\alpha$ with (16) in (13)–(15) and replace $|p_{ij}-p^*_{ij}|\leq\varepsilon$ with $p_{ij}-p^*_{ij}\geq-\varepsilon$ and $p_{ij}-p^*_{ij}\leq\varepsilon$ to remove the two constraints with absolute

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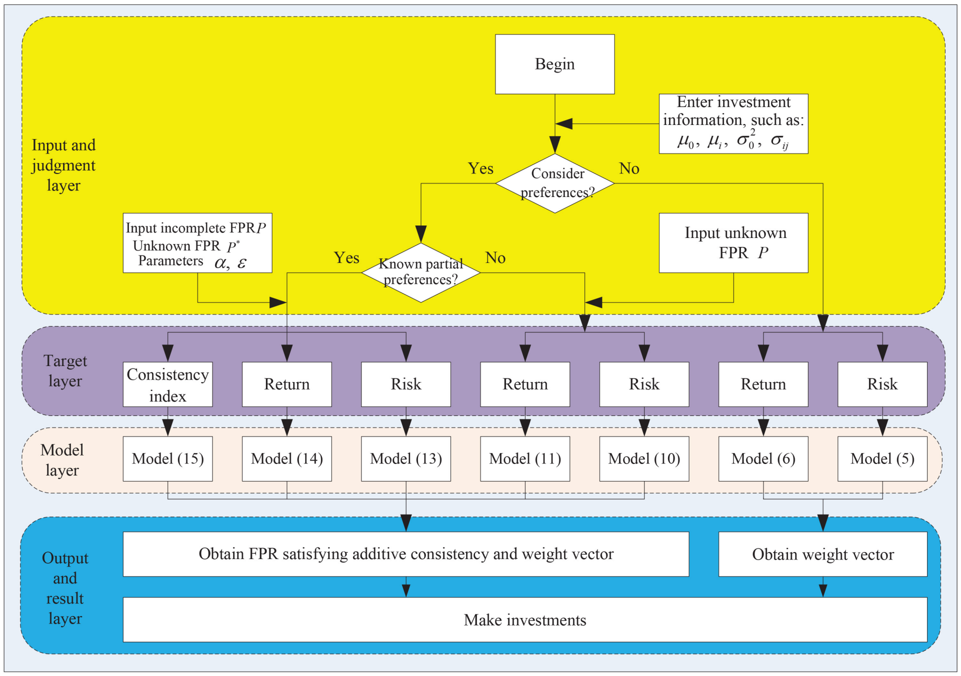

Fig. 1. Investment process.

values. This reduces the computational complexity of these models.

V. Comparative Analysis and Portfolio Process

A. Model Comparison

In Section II, the classical Markowitz portfolio models were reviewed. We then proposed portfolio models with no preference information and portfolio models with incomplete preference information in Section IV; these models have different characteristics, advantages and disadvantages, and adaptations to the environment. This section provides an in-depth comparative analysis of these models.

The main differences between the seven models lie in the objectives and feature constraints—(5), (10), and (13) minimize risk, (6), (11), and (14) maximize returns, and (15) maximizes the consistency index. In terms of advantages and disadvantages, (5) and (6) do not account for decision makers' preference relations for different investment objects, and so both models are suitable for portfolio selection without preference constraints. When decision makers need to satisfy the consistency of their preferences in the investment process, and the preference information is unknown, (10) and (11) return an FPR P that satisfies additive consistency under this investment result. Moreover, when the preference information is partially known, (13)–(15) optimize for risk, return, and the consistency index, respectively, so as to obtain an FPR P with a consistency index no less than α, an FPR P^* that completely satisfies additive consistency, and a priority vector that represents the investment weights at this time.

Previous studies have mainly focused on multiobjective, multistage, multicriteria investments [11], [13], [14] and the form of the rate of return, such as random variables, uncertain variables, or fuzzy numbers [16], [17], [19]. However, the models developed in this article mainly consider the preference information in the portfolio, and connect FPRs with portfolio models for the first time through the priority vector, thus providing a new method for research on behavioral investment.

B. Portfolio Process

Fig. 1 illustrates the investment process under different conditions and circumstances. The investment process is divided into four general steps: 1) the input and judgment layer; 2) the target layer; 3) the model layer; and 4) the output and result layer. When deciding to make an investment, some indispensable information is required, such as μ_0, μ_i, σ_0, and σ_{ij} (note that this information may only be partially required, depending on the situation and the model). In the absence of investment under preferences, the classical models of (5) and (6) can be solved directly. In considering investment under preferences, the investment process can be divided into that with unknown information and that with partially known information. Specifically, when the decision maker's preferences are completely unknown, the FPR P is imported without any data, and (10) and (11) are applied with the objectives of risk and returns, respectively. When the decision maker's preferences are partially known, an incomplete FPR P, an FPR P^* without any data, and the parameters α and ε are required to solve (13)–(15) with the objectives of risk, returns, and the consistency index, respectively. Under the output and result layer, the weight vectors for investment can be obtained from (5) and (6), and the FPR P and the corresponding weight vectors can be obtained from (10) and (11), which completely satisfy additive consistency. Additionally, the FPRs P, P^* and the corresponding weight vectors for P^* can be obtained from (13)–(15). The FPR P is complete and the additive consistency index is not less than α, and

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the FPR P^* completely satisfies additive consistency. Finally, depending on the weight vector, the investment strategy is applied.

VI. Empirical Illustration

In this section, we present an empirical example that illustrates the models and methodology proposed in this article.

A. Background Description and Parameter Assumptions

Suppose that an investor is going to invest in four securities. For convenience, we have selected four securities from each of the four different sectors in the China Stock Market & Accounting Research Database, denoted as A, B, C, D (i.e., $N = 4$, A, B, C, D correspond to $i = 1, 2, 3, 4$), and obtained the daily rates of return for these four securities over the past year. The expected rates of return for these four securities are $\mu = (1.0014, 2.7708, 1.8702, 2.4484)$ (units: 1×10^{-3}), and their covariance matrix $D(r) = (\sigma_{ij})_{n\times n}$ (units: 1×10^{-3}) is

$$(\sigma_{ij})_{4\times 4} = \begin{pmatrix} 0.6259 & 0.0418 & 0.0383 & 0.1077 \\ 0.0418 & 0.9042 & 0.0182 & 0.2175 \\ 0.0383 & 0.0182 & 0.9126 & 0.1178 \\ 0.1077 & 0.2175 & 0.1178 & 1.1100 \end{pmatrix}.$$

Assume that the incomplete FPR given in advance by the investor is

$$P^0 = \begin{pmatrix} 0.5 & 0.73 & ? & ? \\ 0.27 & 0.5 & ? & 0.08 \\ ? & ? & 0.5 & 0.65 \\ ? & 0.92 & 0.35 & 0.5 \end{pmatrix}.$$

We use this information to compare the investment results under the different models.

B. Results Comparison and Sensitivity Analysis for Risk-Minimizing Models

For both Markowitz's portfolio models and the models presented in this article, (5), (10), and (13) are developed to minimize investment risk under the assumption that the rate of return is constant or does not fall below a certain value. However, (10) accounts for the investor's ultimate FPR and produces a completely additively consistent FPR P, whereas (13) considers the investor's prior preferences and not only produces a completely additively consistent FPR P^* but also obtains a complete FPR P with a consistency index no lower than the threshold α. The three models are now compared and analyzed. Setting $\mu_0 = 2, \alpha = 0.8$, and $\varepsilon = 0.2$, the results given by the three models are presented in Table I.

From Table I, the optimal investment weights obtained from (5) and (10) are generally similar because these two models only differ in their equality and inequality constraints relating to the rate of return. In this case, p_{ij} is obtained directly from w_i, w_j. However, the results for (13) differ significantly from those of (5) and (10), particularly in the values of w_2, w_4, because the investor supplies partial preferences before investment. If the optimal investment weights under (5) are adopted, the investor will definitely invest more in security B when w_2 is almost twice w_4. However, because $p_{42} = 0.92$ in the investor's prior FPR P^0, the investor has a very high preference for security D over security B, thereby leading to a much higher outcome for w_4 than w_2 under (13). Although there is a certain amount of increased risk, the investor has minimized their risk without changing their preferences.

TABLE I
Optimal Investment Weights for (5), (10), and (13)

	Model (5)	Model (10)	Model (13)
w_1	0.2756	0.2755	0.2348
w_2	0.3055	0.3056	0.1684
w_3	0.2563	0.2563	0.2817
w_4	0.1626	0.1626	0.3151
$\sum_{i=1}^{n}\sum_{j=1}^{n} w_i\sigma_{ij}w_j$	0.2776	0.2776	0.3128

The FPR $P = (p_{ij})_{n\times n}$ under (10) completely satisfies additive consistency

$$P = \begin{pmatrix} 0.5 & 0.46 & 0.53 & 0.67 \\ 0.54 & 0.5 & 0.57 & 0.71 \\ 0.47 & 0.43 & 0.5 & 0.64 \\ 0.33 & 0.29 & 0.36 & 0.5 \end{pmatrix}.$$

These results for FPR $P = (p_{ij})_{n\times n}$ provide us with values of p_{ij} that satisfy additive consistency under the optimal investment outcomes, thus providing investors with information on their preferences under the optimal investment strategy. Additionally, this information is a reference when formulating the initial preference relation P^0, i.e., any p^0_{ij} that deviates significantly from p_{ij} will inevitably increase the risk for investors, allowing investors to balance their preferences against risk before making a decision.

The results for FPR $P^* = (p^*_{ij})_{n\times n}$ completely satisfy additive consistency under (13), and the complete FPR $P^0 = (p^0_{ij})_{n\times n}$ has a consistency index of no less than 0.8

$$P^* = \begin{pmatrix} 0.5 & 0.6 & 0.43 & 0.38 \\ 0.4 & 0.5 & 0.33 & 0.28 \\ 0.57 & 0.67 & 0.5 & 0.45 \\ 0.62 & 0.72 & 0.55 & 0.5 \end{pmatrix}$$

$$P^0 = \begin{pmatrix} 0.5 & 0.73 & 0.63 & 0.38 \\ 0.27 & 0.5 & 0.53 & 0.08 \\ 0.37 & 0.47 & 0.5 & 0.65 \\ 0.62 & 0.92 & 0.35 & 0.5 \end{pmatrix}.$$

The significance of P^* is that it completely satisfies additive consistency and gives the minimal deviation from the complete P^0. If an investor wishes to increase the consistency index of P^0 without making large changes to their original preferences, P^0_{ij} can be modified in the direction of P^*_{ij}. However, if the investor prefers further risk reduction, P^0 should be made as close as possible to the FPR $P = (p_{ij})_{n\times n}$ that completely satisfies additive consistency under (10). The consequence is that the investor's original preference information may be significantly changed.

The above results are for $\mu_0 = 2, \alpha = 0.8$, and $\varepsilon = 0.2$. Model (13) is the primary model proposed in this article and

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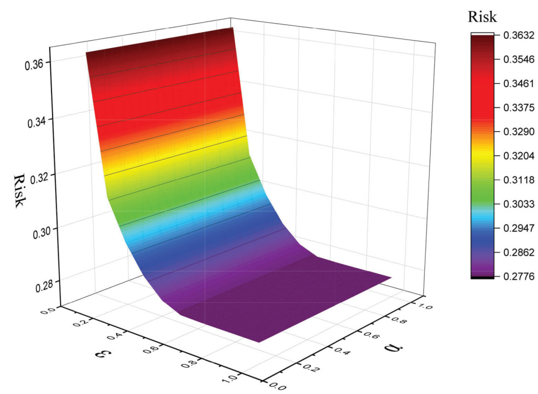

Fig. 2. Minimum risk values under different α and ε in (13).

TABLE II
OPTIMAL INVESTMENT WEIGHTS FOR (6), (11), AND (14)

	Model (6)	Model (11)	Model (14)
w_1	0.2006	0.2006	0.2977
w_2	0.3547	0.3547	0.1474
w_3	0.2569	0.2569	0.2608
w_4	0.1878	0.1878	0.2941
$\sum_{i=1}^{n} w_i\mu_i$	2.1240	2.1240	1.9144

contains all three parameters. Thus, a sensitivity analysis is now performed for (13). In this regard, the minimum risk values under different α and ε are shown in Fig. 2.

Fig. 2 indicates that the minimum risk value under (13) decreases as ε increases, although the rate of decrease progressively slows down. When $\varepsilon = 0.6$, the results are already consistent with those of (5) and (10) because the increase in ε allows investors more flexibility to change their own preferences toward the optimal FPR obtained in (10). However, the minimum risk value under (13) is more robust to α and remains almost stationary under changes in this parameter. This may be because the incomplete FPR given by the decision maker has a large fixed impact on the final complete FPR P^0.

C. Results Comparison and Sensitivity Analysis for Return-Maximizing Models

We now compare (6), (11), and (14), which all seek to maximize returns. The results given by the three models with $\sigma_0^2 = 0.3, \alpha = 0.8$, and $\varepsilon = 0.2$ are presented in Table II.

Table II indicates that the optimal investment weights obtained by (6) and (11) are identical, similar to the case of Table I. This reflects the high similarity between these two models. The fundamental difference is that (10) and (11) obtain the FPR under the optimal solution, while (5) and (6) do not. However, there are significant differences between the results given by (14) and by (6) and (11), especially in the values of w_1, w_2, and w_4. The underlying reason is that $p_{12} = 0.73$ and $p_{42} = 0.92$ in the FPR P^0 previously given by investors. This suggests that, among security classes A, B, and D, investors have a higher preference for security A and a very high preference for security D compared with security B, leading to much higher values for w_1 and w_4 compared with w_2 under (14). Although this reduces the rate of return to some extent, it allows investors to maximize their returns without changing their preferences. Moreover, the FPR $P = (p_{ij})_{n\times n}$ that completely satisfies additive consistency under (11) is as follows:

$$P = \begin{pmatrix} 0.5 & 0.27 & 0.42 & 0.52 \\ 0.73 & 0.5 & 0.65 & 0.75 \\ 0.58 & 0.35 & 0.5 & 0.6 \\ 0.48 & 0.25 & 0.4 & 0.5 \end{pmatrix}.$$

Taken together, this FPR $P = (p_{ij})_{n\times n}$ gives an indication of the values of p_{ij} that satisfy additive consistency for an investment that maximizes the rate of return. By the same token, the FPR provides information about the investor's preferences under the optimal investment and guides the investor when formulating the initial preference relation P^0. For example, a significant deviation from p_{ij} with p_{ij}^0 will inevitably reduce the investor's rate of return. Therefore, we recommend that investors should evaluate their preferences against returns before making a decision.

The FPR $P^* = (p_{ij}^*)_{n\times n}$ that completely satisfies additive consistency under (14) and the complete FPR $P^0 = (p_{ij}^0)_{n\times n}$ with a consistency index of no less than 0.8 are as follows:

$$P^* = \begin{pmatrix} 0.5 & 0.73 & 0.56 & 0.51 \\ 0.27 & 0.5 & 0.33 & 0.28 \\ 0.44 & 0.67 & 0.5 & 0.45 \\ 0.49 & 0.72 & 0.55 & 0.5 \end{pmatrix}$$

$$P^0 = \begin{pmatrix} 0.5 & 0.73 & 0.56 & 0.51 \\ 0.27 & 0.5 & 0.53 & 0.08 \\ 0.44 & 0.47 & 0.5 & 0.65 \\ 0.49 & 0.92 & 0.35 & 0.5 \end{pmatrix}.$$

If the investor expects the rate of return to increase further, then P^0 should be as close as possible to the FPR $P = (p_{ij})_{n\times n}$ that completely satisfies additive consistency under (11). As a consequence, the original information about the preferences of the investor may be significantly affected.

The above results are for $\sigma_0^2 = 0.3, \alpha = 0.8$, and $\varepsilon = 0.2$. Because (14) represents the main model proposed in this article and contains all three parameters, a sensitivity analysis is now performed on (14). Accordingly, the maximum return rates under different α and ε values in (14) are shown in Fig. 3.

From Fig. 3, it can be concluded that higher values of ε increase the rate of return, although the rate of increase gradually becomes saturated. Specifically, when $\varepsilon = 0.7$, the results are consistent with those from (6). Likewise, the rate of return under (14) exhibits a high degree of robustness to α, remaining almost constant under changes in this parameter.

D. Results and Sensitivity Analysis for Consistency-Index-Maximizing Model

To eliminate errors introduced by differences in data, some of the known data from the previous two sections are used to obtain the weight vector $w = (0.2334, 0.1675, 0.2849, 0.3142)$ by substituting $\mu_0 = 2, \sigma_0^2 = 0.3, \varepsilon = 0.2$ into (15) with the objective function $\alpha = 0.9353$. The FPR $P^* = (p_{ij}^*)_{n\times n}$

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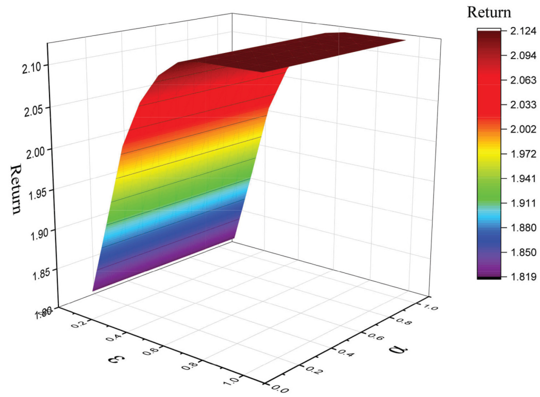

Fig. 3. Maximum returns under different α and ε in (14).

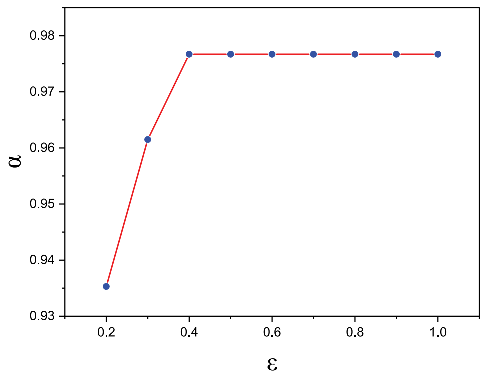

Fig. 4. Maximum consistency index under different ε in (15).

under (15) that completely satisfies additive consistency and the complete FPR $P^0 = (p_{ij}^0)_{n\times n}$ with a consistency index of no less than α are as follows:

$$P^* = \begin{pmatrix} 0.5 & 0.6 & 0.42 & 0.38 \\ 0.4 & 0.5 & 0.33 & 0.28 \\ 0.58 & 0.67 & 0.5 & 0.46 \\ 0.62 & 0.72 & 0.54 & 0.5 \end{pmatrix}$$

$$P^0 = \begin{pmatrix} 0.5 & 0.73 & 0.26 & 0.38 \\ 0.27 & 0.5 & 0.12 & 0.08 \\ 0.74 & 0.88 & 0.5 & 0.65 \\ 0.62 & 0.92 & 0.35 & 0.5 \end{pmatrix}.$$

In this case, the consistency index of the complete P^0 with respect to P^* is not 1, but this is the FPR with the smallest deviation from P^* and the consistency index reaches 0.9353. P^0 preserves all the preference information of the investor and P^* has a priority vector that is the optimal investment weight vector satisfying the return and risk constraints. The maximum consistency index under different ε in (15) with $\mu_0 = 2, \sigma_0^2 = 0.3$ is shown in Fig. 4.

Fig. 4 illustrates that the maximum consistency index in (15) increases as ε increases, and the consistency index is above 0.9353 in all cases. This demonstrates that (15) obtains P^0 with a high additive consistency index. Moreover, when $\varepsilon = 0.4$, the maximum value of α reaches 0.9767, which almost completely satisfies additive consistency. The value of α is not 1 because the initial FPR P^0 given by the investor is incomplete and inconsistent, and the FPR P^0 is derived and its consistency index maximized without changing the known information. Furthermore, from Figs. 2–4, an increase in ε contributes positively to the objective function. Specifically, the risk decreases as ε increases in Fig. 2, the return increases as ε increases in Fig. 3, and the consistency index increases as ε increases in Fig. 4.

VII. Conclusion

During the actual portfolio selection process, investors have different preference relations for different securities, influenced by factors, such as psychology, environment, experience, and data. Thus, the investors' aims are not limited to maximizing returns and minimizing risks under Markowitz's classical models but also to optimize the portfolio process while retaining as much information as possible about their preferences. To the best of our knowledge, this article is the first to introduce FPRs into the portfolio process and develop a series of corresponding models. The main innovations and contributions of this article are as follows.

1) We have improved the relations between the judgment element and the priority vector in FPRs, and provided a new definition of additively consistent FPRs under the priority vector.
2) We proposed two portfolio models considering FPR with unknown preference information, and three portfolio models considering FPR with partially known preference information.
3) We compared the results of all the models developed in this article and performed a sensitivity analysis of the parameters α and ε.
4) Our results reveal that an increase in ε has a positive effect on the objective function of the portfolio models considering FPRs, although this effect gradually becomes saturated. In addition, the models proposed in this article are highly robust to α, basically remaining unchanged with respect to this parameter.

The main limitation of this study is that only the most basic portfolio models and the simplest FPRs have been considered. In future work, complex portfolio selection models with different preference relations, such as interval FPRs, intuitionistic FPRs, hesitant fuzzy linguistic preference relations, and uncertain preference relations, along with their consistency definitions, will be studied. We also hope to explore the mechanisms by which different preference information affects investors, the degree of influence, and the process of achieving optimal portfolios.

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